

D10 boundary-current advances for the de Vries slice

dv2 working note

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Abstract

This note collects the new calculations in the D10 boundary-current route to the de Vries electroweak number. The construction uses Type IIA data on $M_4 \times \mathbb{CP}^2 \times \mathbb{CP}^1$, a D0/M-circle probe side, and a D8-like boundary carrier $\mathcal{W}_9 = M_4 \times \mathbb{CP}^2 \times S_Q^1$. It gives a positive image-channel Feshbach algebra, gives $\kappa_{\text{cur}} = 1$ on the full-current branch, and packages the scalar slice in the coupled variables $\rho, \tau, v_Q, \lambda_H, \kappa_{\text{cur}}$. The new residue calculation shows that the raw Wilson second variation has residue J_j , while the finite Feshbach pole requires unit residue. The unit inverse image metric is the static Schur complement of the retained image oscillator. For a generic local oscillator the pole depends on a residue ratio $r = \gamma/\alpha$. A single current-displacement square forces $r = 1$. The mixed Chan–Paton sector $H^0(\mathbb{CP}^1, \text{Hom}(\mathcal{O}(0), \mathcal{O}(2j)))$ has the right rank: $Q_j = A_j y_j / \sqrt{J_j}$ lies in $\Im A_j$ for $j > 0$. The current D10 boundary package contains no one-square Hom action that would lock its coefficient. The D8/Payen/RR source package also contains no term fixing the full-current placement: $\kappa_{\text{cur}} = 1$ is a branch condition in the written action, while a horizontal-only image channel gives $\sin^2 \theta_W = 0.242265 \dots$. The neutral scalar Hessian has enough coefficient freedom to fit the negative-branch scalar constants; it does not force them in the present action. The source-balance equations reduce any replacement source sector to a two-component lower-charge target; the minimal target is $-h_Q^2$, not a field already present in the current D8/O8 package. A tower/moduli branch instead retains the $j = 3/2$ state, with mass 96.54 GeV when the scale is fixed by M_Z .

1 The de Vries datum

Let

$$X_+(J) = \frac{-J + \sqrt{J^2 + 4J}}{2}, \quad J = j(j+1). \quad (1)$$

The two electroweak slots use

$$J_W = \frac{3}{4}, \quad J_Z = 2. \quad (2)$$

Then

$$\sin^2 \theta_{\text{dV}} = 1 - \frac{X_+(3/4)}{X_+(2)} = 0.2231013223008662, \quad (3)$$

and

$$\frac{M_W}{M_Z} = \sqrt{\frac{X_+(3/4)}{X_+(2)}} = 0.8814185598789793. \quad (4)$$

The D10 calculation below asks whether this number is forced by local boundary dynamics, not only placed in a compatible parameter slice.

2 D10 carrier and probe story

The active ten-dimensional layer is Type IIA on

$$M_4 \times K_6, \quad K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1. \quad (5)$$

The M-circle reduction gives the string-frame form

$$ds_{11}^2 = e^{-2\Phi/3} ds_{10,\text{str}}^2 + e^{4\Phi/3} (d\psi + C_1)^2, \quad F_2 = dC_1. \quad (6)$$

The D0-brane is the electric probe of C_1 :

$$S_{\text{D0,WZ}} = \mu_0 \int_{W_1} C_1, \quad Q_{\text{D0}} \leftrightarrow P_\psi. \quad (7)$$

The space-filling side in Type IIA is the D8-like boundary carrier

$$\mathcal{W}_9 = M_4 \times \mathbb{CP}^2 \times S_Q^1. \quad (8)$$

It is codimension one in K_6 . The literal all-Neumann D9 row is retained only as an open-string control row.

Proposition 2.1 (Carrier/probe separation). *The D0 and $\mathcal{W}_9$ rows test different pieces of one D10 background. The D0 measures the M-circle/RR one-form. The D8-like boundary carries the Q -preserving endpoint/current sector and can support the Chan–Paton representations used by the weak boundary modes.*

Proof. Equation (7) couples the D0 worldline to C_1 , the connection in (6). Equation (8) wraps the observed spacetime, the color $\mathbb{CP}^2$, and the weak Q -circle, leaving one weak normal direction. The worldvolume dimension and RR parity therefore match a Type IIA D8-like source, not a D9-brane. $\square$

3 Boundary action and photon protection

The weak endpoint spaces are

$$V_j = H^0(\mathbb{CP}^1, \mathcal{O}(2j)), \quad j \in \tfrac{1}{2}\mathbb{Z}_{\geq 0}. \quad (9)$$

Borel–Weil gives $T^a T_a = j(j+1)\mathbf{1}_{V_j}$. The finite current branch keeps the level-two set

$$j = 0, \tfrac{1}{2}, 1. \quad (10)$$

The tower branch keeps higher j .

The Q -preserving boundary field is a section

$$\eta \in H^0(\mathbb{CP}^1, \mathcal{O}(1)) \otimes L_Y, \quad Q = T_3 + Y. \quad (11)$$

Choosing the lower doublet component $\eta_0 = v$ gives

$$T_3 \eta_0 = -\tfrac{1}{2} \eta_0, \quad Y \eta_0 = \tfrac{1}{2} \eta_0, \quad Q \eta_0 = 0. \quad (12)$$

Proposition 3.1 (Photon protection). *On the $\eta = \eta_0$ boundary slice, the Q -connection has no mass term from $\langle D\eta, D\eta \rangle$.*

Proof. The contribution proportional to the Q -connection is $iA_Q Q \eta_0$, which vanishes by (12). Generators that do not annihilate η_0 contribute quadratic terms $\langle R\eta_0, R\eta_0 \rangle A_R^2$. $\square$

4 Image-channel Feshbach algebra

Let e_a , $a = 1, 2, 3$, be an orthonormal weak-current frame. Define

$$A_j : V_j \longrightarrow V_j \otimes \mathbb{R}^3, \quad A_j p = \sum_{a=1}^3 T_a^{(j)} p \otimes e_a. \quad (13)$$

Then

$$A_j^\dagger A_j = J_j \mathbf{1}_{V_j}, \quad J_j = j(j+1). \quad (14)$$

For $j > 0$, let $q_j \in \text{Im } A_j$ and

$$P_j = \frac{1}{J_j} A_j A_j^\dagger \quad (15)$$

be the image projector.

The positive image-channel quadratic form is

$$S_{\text{quad}}^{(j)} = \langle p_j, B p_j \rangle + \langle q_j, (B + \mu^2 J_j) q_j \rangle - \mu^2 \langle q_j, A_j p_j \rangle - \mu^2 \langle A_j p_j, q_j \rangle. \quad (16)$$

Eliminating q_j gives

$$\Gamma_j(B) = B - \mu^4 A_j^\dagger (B + \mu^2 J_j)^{-1} A_j = B - \frac{\mu^4 J_j}{B + \mu^2 J_j}. \quad (17)$$

With $X = B/\mu^2$, the pole equation is

$$X = \frac{J_j}{X + J_j}, \quad (18)$$

which is (1).

5 Canonical current metric

Let K_a be the weak Killing/current generators in the same internal metric that appears in the Kaluza–Klein reduction. Define

$$G_{ab}^{(2)} = \frac{1}{\text{Vol}(\widehat{K}_6)} \int_{\widehat{K}_6} \sqrt{g} g_{mn} K_a^m K_b^n. \quad (19)$$

On the $SU(2)$ -symmetric weak line,

$$G_{ab}^{(2)} = L_2^2 \delta_{ab}, \quad e_a = \frac{K_a}{L_2}. \quad (20)$$

In this frame (14) gives

$$\kappa_{\text{cur}} = 1. \quad (21)$$

This conclusion is conditional on the boundary image channel using the full current $K_{a,\text{full}}$. The Bailin–Love split has

$$L_{2,BL}^2 = L_{2,\text{horiz}}^2 + L_{2,\text{vert}}^2 = (1 + \beta) L_{2,\text{horiz}}^2. \quad (22)$$

If the endpoint image channel sees only the horizontal $\mathbb{CP}^1$ current, then

$$\kappa_h = \frac{L_{2,\text{horiz}}^2}{L_{2,BL}^2} = \frac{1}{1 + \beta} = 0.7527390090748721, \quad (23)$$

and the deformed pole gives

$$\sin^2 \theta_W = 0.24226537471822063. \quad (24)$$

The Payen endpoint action supplies the representation tensor for the connection inserted into the boundary Wilson loop. The D8 WZ term supplies RR charge data. Neither term in the current source package fixes the placement of $L_{2,vert}$ inside the endpoint current metric.

Proposition 5.1 (Full-current normalization). *The boundary image map and the weak gauge metric share the same Casimir normalization on the full-current branch. A horizontal-only boundary current changes the normalization to*

$$\kappa_{\text{cur}} = 0.7527390090748721, \quad \sin^2 \theta = 0.24226537471822063.$$

Proof. Equation (20) orthonormalizes the same current vectors used in the Kaluza–Klein gauge metric. Hence

$$A_j^\dagger A_j = \sum_{a,b} T_a^{(j)} T_b^{(j)} \langle e_a, e_b \rangle = \sum_a T_a^{(j)} T_a^{(j)} = J_j \mathbf{1}_{V_j}.$$

Replacing the boundary image metric by only the horizontal component rescales the current frame by the recorded factor, which shifts the weak-angle output. $\square$

6 Wilson-contact residue

The static part of (16) is

$$S_{\text{stat}} = \mu^2 J_j \|q_j\|^2 - \mu^2 \langle q_j, A_j p_j \rangle - \mu^2 \langle A_j p_j, q_j \rangle. \quad (25)$$

Completing the square in $\text{Im } A_j$ gives

$$S_{\text{stat}} = \mu^2 J_j \left\| q_j - \frac{A_j p_j}{J_j} \right\|^2 - \mu^2 \|p_j\|^2. \quad (26)$$

Thus the finite image-channel pole fixes the open-channel contact as

$$C_{\text{req}} = -\mu^2 \mathbf{1}_{V_j}. \quad (27)$$

The Wilson factor expansion gives

$$\left. \frac{\delta^2 W_j}{\delta \xi^a \delta \xi^b} \right|_{\xi=0} \sim -\mu^2 T_a^{(j)} T_b^{(j)}. \quad (28)$$

In the orthonormal current frame this contracts to

$$C_{\text{Wilson}} = -\mu^2 \sum_a T_a^{(j)} T_a^{(j)} = -\mu^2 J_j \mathbf{1}_{V_j}. \quad (29)$$

Projection to the image does not change the coefficient:

$$A_j^\dagger P_j A_j = A_j^\dagger A_j = J_j \mathbf{1}_{V_j}. \quad (30)$$

Proposition 6.1 (Residue test). *The raw Wilson second variation has residue J_j , while the finite Feshbach pole requires unit residue. The unit contact is*

$$-\mu^2 A_j^\dagger \left(A_j A_j^\dagger|_{\text{Im } A_j} \right)^{-1} A_j = -\mu^2 \mathbf{1}_{V_j}. \quad (31)$$

Proof. Equations (26) and (27) fix the contact required by the pole. Equations (29) and (30) compute the contact from the Wilson Taylor term and from the projected image tensor. On $\text{Im } A_j$, $A_j A_j^\dagger = J_j \mathbf{1}$, so

$$A_j^\dagger \left(A_j A_j^\dagger|_{\text{Im } A_j} \right)^{-1} A_j = \frac{1}{J_j} A_j^\dagger A_j = \mathbf{1}_{V_j}.$$

□

If the inverse image metric is absent and the Wilson residue is used, the kernel becomes

$$\Gamma_j(B) = B + \mu^2(1 - J_j) - \frac{\mu^4 J_j}{B + \mu^2 J_j}. \quad (32)$$

The pole equation is

$$X + 1 - J_j - \frac{J_j}{X + J_j} = 0. \quad (33)$$

This changes the W and Z slots by different amounts and cannot be absorbed into one common scale.

The inverse in (31) is not a separate microscopic operator when q_j is retained. It is the $B = 0$ Schur complement of the local image oscillator:

$$\Gamma_j(0) = -\mu^2 A_j^\dagger \left(A_j A_j^\dagger|_{\text{Im } A_j} \right)^{-1} A_j = -\mu^2 \mathbf{1}_{V_j}. \quad (34)$$

The finite branch is shifted only if the raw Wilson $T_a T_b$ term is also kept as a direct 1PI p_j contact.

The endpoint action is linear in the source. Therefore the Wilson second derivative is a connected two-current insertion rather than a microscopic quadratic seagull. If q_j is retained as the local image-current channel, the connected response is represented by q_j exchange and should not be added again as a direct 1PI p_j contact.

Coefficient lock

For a canonically normalized image oscillator, the most general single-pole quadratic form with the same tensor structure is

$$\begin{aligned} S_j^{(2)} = & \langle p_j, B p_j \rangle + \langle Q_j, (B + \alpha \mu^2 A_j A_j^\dagger) Q_j \rangle \\ & - \gamma \mu^2 \langle Q_j, A_j p_j \rangle - \gamma \mu^2 \langle A_j p_j, Q_j \rangle. \end{aligned} \quad (35)$$

Eliminating Q_j gives

$$\Gamma_j(B) = B - \frac{\gamma^2 \mu^4 J_j}{B + \alpha \mu^2 J_j}. \quad (36)$$

With $\mu_{\text{eff}}^2 = \alpha \mu^2$ and $X = B/\mu_{\text{eff}}^2$, the pole equation is

$$X = \frac{r^2 J_j}{X + J_j}, \quad r = \frac{\gamma}{\alpha}. \quad (37)$$

The de Vries pole requires $r = 1$.

The local square

$$\alpha \mu^2 \|A_j^\dagger Q_j - p_j\|^2 - \alpha \mu^2 \|p_j\|^2 \quad (38)$$

expands to (35) with $\gamma = \alpha$. Therefore a single current-displacement square gives the needed coefficient lock; a generic oscillator does not.

DBI/Stueckelberg displacement terms have this one-square form, so they provide the right coefficient mechanism. The present D8 normal displacement is only a scalar on $\mathcal{W}_9$, however; it is not a field in $\Im A_j \subset V_j \otimes \mathbb{R}^3$. Thus the finite branch still requires an $\Im A_j$ vector-bundle displacement supplied by the D8/open-current interface.

The mixed Chan–Paton boundary-changing sector supplies a sharper candidate. For $j > 0$,

$$H^0(\mathbb{CP}^1, \text{Hom}(\mathcal{O}(0), \mathcal{O}(2j))) = V_j, \quad U_j = \frac{1}{\sqrt{J_j}} A_j : V_j \rightarrow \Im A_j \quad (39)$$

is an isometry. Thus a mixed zero mode $y_j \in V_j$ gives a canonical image field

$$Q_j = U_j y_j = \frac{1}{\sqrt{J_j}} A_j y_j, \quad A_j^\dagger Q_j = \sqrt{J_j} y_j. \quad (40)$$

The desired current-displacement square is equivalent to

$$\alpha \mu^2 \|\sqrt{J_j} y_j - p_j\|^2 - \alpha \mu^2 \|p_j\|^2. \quad (41)$$

This passes the field-content test but not the local-action test. A generic Hom-mode quadratic form

$$\begin{aligned} S_y^{(2)} = & \langle p_j, B p_j \rangle + Z_y \langle y_j, B y_j \rangle + c_m \mu^2 J_j \langle y_j, y_j \rangle \\ & - c_g \mu^2 \sqrt{J_j} (\langle y_j, p_j \rangle + \langle p_j, y_j \rangle) \end{aligned} \quad (42)$$

gives

$$\Gamma_j(B) = B - \frac{c_g^2 \mu^4 J_j}{Z_y B + c_m \mu^2 J_j}, \quad r = \frac{c_g \sqrt{Z_y}}{c_m}. \quad (43)$$

The de Vries pole requires $c_m = c_g \sqrt{Z_y}$. The mixed zero-mode statement supplies the bundle; it does not by itself supply this coefficient lock.

The current source package contains no term of the form (41). The Payen endpoint action gives the representation, Wilson loop, and charge insertions $T_a, T_a T_b$. It is first order on the boundary and contains no propagating Hom-mode potential. The D8 DBI/WZ package supplies geometric brane norms and RR charge data, but no term identifying the mixed Hom mode with the mismatch $\sqrt{J_j} y_j - p_j$. With the current action, the Hom sector therefore has the generic ratio (43).

7 Finite branch and tower branch

The residue calculation separates two branches.

Branch	Spectrum rule	Phenomenological signal
Finite current branch	The level-two endpoint/current sector keeps $j = 0, \frac{1}{2}, 1$. The Feshbach pole uses (31).	No $j = 3/2$ state in the retained weak boundary spectrum.
Tower/moduli branch	The rotor or neighboring moduli slice keeps higher j . The natural gap is proportional to $j(j+1)$.	The first extra positive branch is $j = 3/2$, near the low-mass diphoton window.

For $j = 3/2$, $J = 15/4$ and

$$X_+\left(\frac{15}{4}\right) = 0.8204823316059779. \quad (44)$$

Fixing the common scale by $M_Z = 91.1880 \text{ GeV}$ gives

$$M_{3/2} = M_Z \sqrt{\frac{X_+(15/4)}{X_+(2)}} = 96.53875533197167 \text{ GeV}. \quad (45)$$

Thus a tower/moduli branch places a neutral $j = 3/2$ mass inside the CMS Run 2 low-mass diphoton search range and 1.14 GeV above the reported 95.4 GeV local feature.

8 Extra-mode gap check

The finite image operator is safe only on the image itself. If the field is an unprojected element of $V_j \otimes \mathbb{R}^3$, then

$$V_j \otimes \mathbb{R}^3 = \Im A_j \oplus (\Im A_j)^\perp.$$

On $\Im A_j$, $A_j A_j^\dagger = J_j$. On the complement,

$$A_j^\dagger q_\perp = 0, \quad A_j A_j^\dagger q_\perp = 0.$$

Thus the quadratic operator $B + \mu^2 A_j A_j^\dagger$ gives denominator B on $q_\perp$. The complement is massless unless the action supplies the projection $q_j = P_j q_j$ or a positive complement mass.

The mixed Hom field avoids this complement because

$$y_j \in H^0(\mathbb{CP}^1, \text{Hom}(\mathcal{O}(0), \mathcal{O}(2j))) = V_j, \quad Q_j = \frac{1}{\sqrt{J_j}} A_j y_j \in \Im A_j.$$

This fixes the finite field content, but the current D8/Payen/RR package contains no one-square Hom action.

The rotor branch has the opposite problem. With the Z fixing the scale, the next tower masses are

j	$M_j \text{ GeV}$
3/2	96.5388
2	99.5795
5/2	101.4539
3	102.6807

and the large- j limit is 106.57 GeV. The rotor branch is therefore not a high-gap branch.

9 Coupled selector potential

The boundary scale can be coupled to closed-string moduli by writing $\mu_B = \mu_B(\rho, \tau)$. The Type IIA scaling potential is

$$V_{\text{bulk}} = -B_R \tau^{-2} \rho^{-2} + B_H \tau^{-2} \rho^{-6} + B_0 \tau^{-4} \rho^6 + B_2 \tau^{-4} \rho^2 \\ + B_4 \tau^{-4} \rho^{-2} + B_6 \tau^{-4} \rho^{-6} + B_{\text{loc}} \tau^{-3}. \quad (46)$$

Set

$$J_v = 1 + \sqrt{3}, \quad J_H = \frac{3 + \sqrt{57}}{8}, \quad \lambda_{\text{dV}} = \frac{J_H}{4J_v} = 0.12067210653215474. \quad (47)$$

The coupled selector potential is

$$\begin{aligned}
V_{\text{sel}} = & C_{\text{cur}}(\rho, \tau)(\kappa_{\text{cur}} - 1)^2 \\
& + C_v(\rho, \tau)(v_Q^2 - \mu_B(\rho, \tau)^2 J_v)^2 \\
& + C_H(\rho, \tau)(\lambda_H - \lambda_{\text{dV}})^2, \quad C_{\text{cur}}, C_v, C_H > 0.
\end{aligned} \tag{48}$$

At the radial minimum $h^2 = 2v_Q^2$, stationarity in the selector variables gives

$$\kappa_{\text{cur}} = 1, \quad v_Q^2 = \mu_B(\rho, \tau)^2(1 + \sqrt{3}), \quad \lambda_H = \lambda_{\text{dV}}. \tag{49}$$

With $v_{\text{EW}} = \sqrt{2}v_Q$,

$$M_H^2 = 2\lambda_H v_{\text{EW}}^2. \tag{50}$$

On the selector branch the squared selector terms vanish, so the ρ, τ stationarity equations reduce to the bulk coefficient balances from (46).

The selector terms do not select ρ, τ on that branch. If

$$S_{\text{cur}} = \kappa_{\text{cur}} - 1, \quad S_v = v_Q^2 - \mu_B(\rho, \tau)^2 J_v, \quad S_H = \lambda_H - \lambda_{\text{dV}},$$

then $V_{\text{sel}} = \sum_i C_i(\rho, \tau) S_i^2$, and at $S_i = 0$

$$\partial_\rho V_{\text{sel}} = \sum_i (\partial_\rho C_i) S_i^2 + 2 \sum_i C_i S_i \partial_\rho S_i = 0$$

with the same statement for ∂_τ . The bulk equations are therefore the only ρ, τ stationarity equations in this written potential.

Those bulk equations are two linear constraints on the seven coefficients

$$B_R, B_H, B_0, B_2, B_4, B_6, B_{\text{loc}}.$$

For instance, in the reduced coefficient slice

$$B_H = B_0 = B_2 = B_6 = 0,$$

stationarity at any chosen ρ is obtained by

$$B_4 = B_R \tau^2, \quad B_{\text{loc}} = -\frac{2}{3} B_R \tau \rho^{-2}.$$

At $\rho_* = 9.463469757158872$, this gives

$$B_{\text{loc}} = -0.007444027070759169 B_R \tau.$$

Thus the written vacuum potential supplies a coefficient-balance parametrization, not a selection of ρ_* . The source-balance obstruction below also removes the current D8/O8 localized source package as a global completion for B_{loc} .

10 Neutral scalar Hessian

The scalar constants in (47) are the absolute values of the negative root of the same quadratic as (1):

$$|X_-(2)| = 1 + \sqrt{3} = J_v, \quad |X_-(3/4)| = \frac{3 + \sqrt{57}}{8} = J_H. \quad (51)$$

This extends the de Vries algebra to the scalar slice. The local scalar Hessian test asks whether the boundary action forces this extension.

For a canonically normalized neutral radial field h , write

$$V_h(h) = a(\rho, \tau)h^2 + b(\rho, \tau)h^4 + V_0(\rho, \tau), \quad b > 0. \quad (52)$$

At a nonzero stationary point,

$$h_0^2 = -\frac{a}{2b}, \quad M_h^2 = V_h''(h_0) = 8bh_0^2 = -4a. \quad (53)$$

The selector normalization is $h_0^2 = v_{\text{EW}}^2 = 2v_Q^2$. If $v_Q^2 = J_v\mu_B^2$, then

$$\frac{M_h^2}{\mu_B^2} = 16bJ_v. \quad (54)$$

Equivalently,

$$V_h = \frac{\lambda_H}{4}(h^2 - 2v_Q^2)^2 \Rightarrow M_h^2 = 4\lambda_H v_Q^2. \quad (55)$$

The scalar target $M_h^2 = J_H\mu_B^2$ is obtained by setting

$$\lambda_H = \frac{J_H}{4J_v}. \quad (56)$$

Thus the scalar formula is internally consistent, but the coefficient b , or equivalently λ_H , is a free scalar coefficient in the present local potential.

Adding neutral scalar mixing does not change this rank count. For neutral variables $\Phi^A = (h, \phi^i)$, the physical scalar masses are eigenvalues of

$$H_{AB} = \frac{\partial^2 V}{\partial \Phi^A \partial \Phi^B}. \quad (57)$$

Requiring one eigenvalue to be $J_H\mu_B^2$ is one spectral condition on the entries of H_{AB} . Positivity restricts signs, but it does not select the square-root value J_H . The current scalar sector therefore realizes the negative branch as a compatible target, not as a forced Hessian eigenvalue.

11 Source balance

The D8/O8 WZ bookkeeping gives a compact source table. Let

$$\frac{F_Q}{2\pi} = f_m h_Q, \quad f_m = \frac{2m + 13}{2}. \quad (58)$$

The D8 stack contributes

$$Q_{D8} = N_8, \quad Q_{D6} = f_m N_8 h_Q, \quad Q_{D4} = \frac{2(2m + 13)^2 - 1}{16} N_8 h_Q^2. \quad (59)$$

The local O8 contribution is

$$Q_{O8} = -16 - \frac{1}{2}h_Q^2. \quad (60)$$

Spin-c integrality requires $N_8 \in 16\mathbb{Z}$. In a two-endpoint type I' bookkeeping with total stack $N_{8,L} + N_{8,R} = 32$, the total eight-charge cancels, but the induced lower vector is

$$Q_{\text{lower}}^{\text{total}} = 32f_m h_Q + [2(2(2m+13)^2 - 1) - 1] h_Q^2. \quad (61)$$

For the minimal spin-c lift,

$$m = 0 : \quad Q_{\text{lower}}^{\text{total}} = 208h_Q + 673h_Q^2. \quad (62)$$

Thus the current source package does not pass the global tadpole test. The eight-charge can be balanced, but no current bulk flux, image sector, K-theory identification, or extra localized source cancels (62).

More generally, let an added lower-source sector contribute

$$Q_{\text{new}} = \alpha h_Q + \beta h_Q^2. \quad (63)$$

The lower Bianchi equations require

$$\alpha = -\frac{N_L M_L + N_R M_R}{2}, \quad (64)$$

and

$$\beta = -\left[\frac{(2M_L^2 - 1)N_L + (2M_R^2 - 1)N_R}{16} - 1 \right]. \quad (65)$$

For an unequal endpoint stack this is

$$Q_{\text{new}} = -16M h_Q - (4M^2 - 3)h_Q^2, \quad M \in 2\mathbb{Z} + 1. \quad (66)$$

For a balanced endpoint stack with $M_R = -M_L$, the D6 entry vanishes and

$$Q_{\text{new}} = -(4M_L^2 - 3)h_Q^2. \quad (67)$$

The minimal balanced choice $M_L = \pm 1$ therefore needs

$$Q_{\text{new}} = -h_Q^2. \quad (68)$$

This obstruction is not removed by changing the two endpoint spin-c lifts inside the present type I' data. Write $M_i = 2m_i + 13$. Unequal stack distributions leave

$$Q_6 = 16M_i h_Q, \quad (69)$$

which cannot vanish because M_i is odd. Balanced stacks can cancel D6 only when $M_R = -M_L$. Then

$$Q_4 = (4M_L^2 - 3)h_Q^2, \quad (70)$$

whose minimum is h_Q^2 . Thus the endpoint data alone cannot cancel all lower charges.

Closed H -flux does not remove the obstruction on the selected spaces. The product and flag cohomology rings have only even generators, hence

$$H^3 = H^5 = 0. \quad (71)$$

With no NS5 source, $[H] = 0$, so the cohomology classes $[HF_0]$ and $[HF_2]$ vanish. The existing NSNS flux entry in the potential scaffold is therefore not a cancellation of the lower source vector.

Nor can the current lower vector be removed by a bare K-theory identification. For the product and flag spaces the cohomology is torsion-free and concentrated in even degree, so the rational Chern character detects the lower charge entries. A nonzero rational h_Q or h_Q^2 component is not K-theory equivalent to zero in the charge group used by the WZ action.

The lower-only source requirement can be isolated. In the same-lift source table, an added image sector with no D8 charge would need

$$Q_{\text{img}} = -208h_Q - 673h_Q^2. \quad (72)$$

The balanced opposite-lift endpoint package cancels the D6 part first and leaves the residual D4 source

$$Q_4 = (4M_L^2 - 3)h_Q^2, \quad M_L \in 2\mathbb{Z} + 1. \quad (73)$$

The smallest residual is h_Q^2 . Hence the minimal lower-only completion is a pure negative D4 image/source

$$Q_{\text{img}} = -h_Q^2. \quad (74)$$

A literal anti-D4 or an O4-like image source has this charge type, but either choice is an enlargement of the source package and brings additional localized tension, backreaction, and potential coefficients.

The O4 normalization is favorable. The orientifold WZ coefficient -2^{p-5} gives an $O4^-$ charge -1 in the doubled type I' convention. Thus an $O4^-$ fixed component on $\{pt\}_{\mathbb{CP}^2} \times S_Q^1$ would supply $-h_Q^2$. The product fixed-locus geometry does not give this component by itself. A holomorphic projective involution on $\mathbb{CP}^2$ fixes $\mathbb{CP}^0 \sqcup \mathbb{CP}^1$, so with the RP^1 fixed circle in $\mathbb{CP}^1$ it produces O4 and O6 supports. Standard complex conjugation fixes RP^2 and therefore gives O6-type support after the $\mathbb{CP}^1$ fixed circle is included. The O4 charge match therefore requires an enlarged O4/O6 source table rather than closing the present D8/O8 package.

The enlarged holomorphic package still does not close the source table. In the same doubled convention the companion $O6^-$ carries $-4h_Q$, so one product involution contributes

$$Q_{O4/O6} = -4h_Q - h_Q^2. \quad (75)$$

The D8 endpoint D6 charge is

$$Q_6^{D8} = \frac{N_L M_L + N_R M_R}{2} h_Q, \quad N_i \in \{0, 16, 32\}, \quad M_i \in 2\mathbb{Z} + 1, \quad (76)$$

hence $Q_6^{D8} \in 16\mathbb{Z} h_Q$. It cannot cancel the single O6 unit. With k identical O4/O6 packages, D6 cancellation forces k to be a multiple of four, while the balanced endpoint D4 coefficient $4M^2 - 3$ is 1 modulo 8. Thus the O4/O6 enlargement does not close Milestone 6 inside the current endpoint lattice.

The source branch is therefore closed for the present compactification ansatz. A standalone anti-D4 would cancel the minimal cohomology class, but it adds localized tension, an open sector, and backreaction not included in the boundary-current action. An O4-only repair would require a different orientifold or topology. A D6-bearing repair adds a new source sector and changes the Bianchi table. Each option is a new compactification ansatz, not a completion of the current D8/O8 route.

Proposition 11.1 (Source-balance obstruction). *For the current D8/O8 endpoint package on the selected product or flag topology, with closed H , no NS5 source, rational WZ/K-theory charge detection, and only the tested product O4/O6 enlargement, the lower Bianchi classes cannot vanish.*

Proof. Endpoint D8/O8 data alone fail as follows. Unequal endpoint stacks leave $Q_6 = 16M_i h_Q$, which is nonzero for odd M_i . Balanced stacks cancel D6 only with $M_R = -M_L$, and then leave $(4M_L^2 - 3)h_Q^2$, which is nonzero. Closed H has no cohomology class on the selected spaces because $H^3 = H^5 = 0$, so the HF_0 and HF_2 terms cannot cancel these classes. The rational Chern character detects the nonzero h_Q and h_Q^2 entries on the torsion-free even cohomology spaces, so bare K-theory cannot identify the lower vector with zero. The tested holomorphic O4/O6 package contributes $-4kh_Q - kh_Q^2$. D6 cancellation forces k to be a multiple of four, but the balanced endpoint D4 coefficient is 1 modulo 8, and the unequal endpoint D4 equation $4M_i^2 - 3 = 4M_i$ has no integer solution. Thus all tested cancellation routes fail under the stated assumptions. $\square$

Proposition 11.2. *Under the scoped assumptions above, the de Vries weak-angle relation is not forced by the D10 dynamics. The construction is a compatibility framework: it contains the D10 carrier/probe geometry, the endpoint Casimirs, the finite image-channel algebra, and the full-current metric branch, but at least one mandatory pass condition fails on every route that would turn the algebraic match into a prediction.*

Proof. Milestone 1 fails as a forcing statement because the scoped action does not supply the one-square Hom/image term needed to lock the Feshbach residue. Milestone 2 selects the finite set only on an imposed finite branch; Borel-Weil alone gives a tower. Milestone 3 gives $\kappa_{\text{cur}} = 1$ only on the full-current branch, while the scoped source package does not select that branch. Milestones 4 and 5 leave the bulk coefficients, boundary scale, and scalar Hessian as coefficient targets. Milestone 6 fails global source balance by the preceding proposition and closes the current source branch. Milestone 7 therefore has no derived source/vacuum system from which to compute thresholds, scheme conversion, or branching. Thus the de Vries number is compatible with the scoped route but is not a prediction of it. $\square$

12 Status of the construction

Proposition 12.1 (Current result). *The D10 boundary-current construction gives a coherent compatibility framework for the de Vries weak-angle number. The canonical current metric gives $\kappa_{\text{cur}} = 1$ on the full-current branch, but the written D8/Payen/RR source package contains no term fixing that branch. The finite image-channel Feshbach algebra gives the de Vries pole only when the open-channel contact has unit residue. The raw Wilson contact has residue J_j , so ordinary Wilson expansion plus image projection does not force the finite pole when treated as a separate direct 1PI term. In the retained-field action the Wilson second derivative is a connected response represented by q_j exchange. The local oscillator must also satisfy the coefficient lock $r = 1$ in (37). The mixed Hom sector supplies the right finite image field through (40), but Payen plus the D8 DBI/WZ source package contains no one-square Hom action (41). The source table leaves the lower charge vector (62) uncanceled; after the best endpoint-lift cancellation it still needs a negative D4 image/source $-h_Q^2$. The O4⁻ charge magnitude matches this unit, but the product involutions add O6-type support. The resulting O6 charge is incompatible with the current D8 endpoint lattice, so the source package is not closed. The scoped source-balance obstruction above rules out the current D8/O8 package as a global compactification proof. The extra-mode check also fails for the current action: an unprojected image field has a massless complement, while the rotor branch*

keeps a light tower. The vacuum rank test shows that the written selector and bulk potential do not select ρ_ . The scalar Hessian rank test shows that the negative-branch scalar constants are fitted by free scalar coefficients in the present action. The replacement-source equations (64)–(68) reduce the minimal source repair to $-h_Q^2$, which is not supplied by the current package. Accepting that repair changes the source content or topology, so the current compactification branch is not a proof branch.*

Proof. The compatibility framework is the collection of (5), (8), (9), (12), (16), (21), and (48). The Feshbach pole follows from (17). The current metric follows from (19) and (20). The residue mismatch follows from comparing (27) with (29). $\square$

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